

# MEHUL BHARDWAJ

Toronto, ON, Canada

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## Education

University of Toronto

Sep. 2020 – May 2025

BASc in Engineering Science - Machine Intelligence Major, Business Minor

## Technical Skills

**Languages:** Python, R, SQL, C, C++, HTML/CSS, JavaScript, MATLAB, UNIX, Git

**Machine Learning:** TensorFlow, PyTorch, GAN, Transformer, RNN, CNN, Deep Learning, Active Learning

**Project Management:** MS Project, Excel, JIRA, PowerBI, Visio, Smartsheet, Confluence

**Soft Skills:** Problem-solving, Adaptability, Communication, Teamwork, Leadership, Time Management, Conflict Resolution

## Work Experience

Data Scientist Intern, Advanced Micro Devices (AMD)

May 2024 – Sept 2024

*Data Analytics & Process Optimization*

- Leveraged cutting-edge Python libraries for advanced data wrangling, ETL, and **Big Data** processing—filtering a massive 500+ column, 300M+ row dataset to drive scalable analytics.
- Engineered key features and organized **cloud** data from **Snowflake** into 10 specialized datasets using **Pandas**, enhancing data quality for **EDA**.
- Leveraged correlation analysis, **SHAP** values, and **hypothesis testing**—to engineer **XGBoost** based predictive model achieving **92%** accuracy and fueling data-driven decision-making.
- Designed an **anomaly** detection pipeline employing interactive data visualization and **k-means** clustering to identify 5 critical variables impacting performance

Project Manager Intern, Advanced Micro Devices (AMD)

May 2023 – Sept 2024

*Supply Chain Management & Dashboard Development*

- Streamlined **workflow** documentation, including detailed **flowcharts** for process **visualization** and **Gantt** charts for timeline tracking, streamlining the onboarding of 10+ new team members.
- Developed **Python** scripts to reduce processing time from over 10 minutes to under 5 seconds, achieving **automated data processing** and minimizing manual errors.
- Deployed an interactive real-time **supply-demand** analysis **dashboard**, optimizing tracking for 200+ **NPI** builds across multiple data center programs.

## Projects

Undergraduate Research Assistant, University of Toronto

Sept 2024 – Present

*Gaussian Process Regression for  $Li_2S$  Manufacturing*

- Developed a **Gaussian Process Regression (GPR)** model with **Active Learning** to optimize lithium sulfide manufacturing and reduce experimental costs.
- Designed interactive **3D visualizations** to map uncertainty and guide **Bayesian Optimization** using acquisition functions like Expected Improvement.
- Applied imputation, uncertainty-aware modeling, and **feature engineering** in a **low-data** regime with significant missing values.

AI/ML Engineer – Capstone Project, GEI Consultants Inc.

Sept 2024 – Feb 2025

*RFP Response System Development*

- Developed an AI-driven **RFP** response system using Large Language Models (**LLMs**) via the OpenAI **API** to streamline proposal generation.
- Implemented Retrieval-Augmented Generation (**RAG**) with Hugging Face **embeddings** to enhance document relevance across 200+ documents.
- Developed custom Python functions to prevent hallucinations by retrieving only **relevant documents** with verifiable references, using **similarity**-based retrieval to enhance accuracy and contextual relevance.

## Extracurricular Activities

Photography Content Creator

*Instagram @photography\_mehul*

- Nature lover capturing the beauty of wildlife and landscapes through photography.